

Reconfigurable DDR5 Memory Controller

MC Core — Design Specification

Sub-Blocks · FSMs · Timing Constraints · DFI 5.2 Interface

Standard	JEDEC JESD79-5, DFI 5.2, AXI4	Burst	BL16 (BL/2 = 8 clk)	FSMs	23 total
Banks	16 (2BG × 4 banks)	Clock	Dual: AXI / MC (CDC at one point)	DFI ratio	1:1, 1:2, 1:4

1 System Overview

The MC Core sits entirely in the MC clock domain, receiving commands from the Client Interface (CIF) via a single Async FIFO CDC bar, and emitting DFI5.2 commands to the DDR PHY. Every DRAM timing constraint, scheduling decision, refresh housekeeping, and power-state transition is the MC Core’s responsibility. It has no knowledge of the AXI protocol — that is fully resolved by the CIF upstream.

1.1 Top-Level Architecture

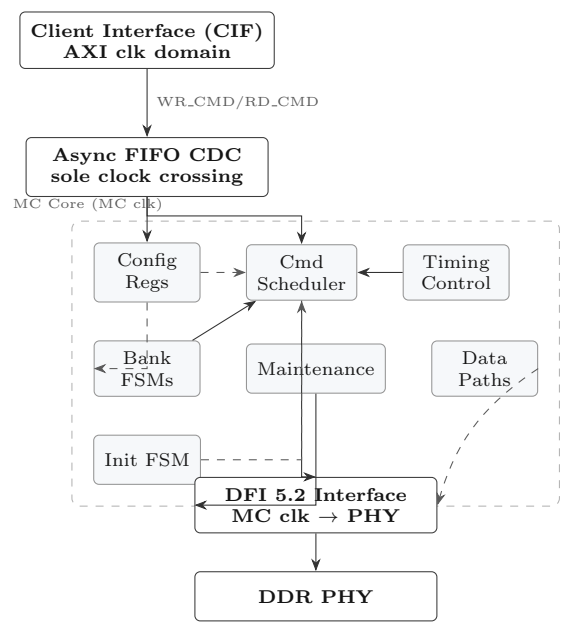


Figure 1. MC Core top-level architecture. The CDC bar is the sole clock-domain crossing. All blocks below it run in the MC clock domain.

1.2 Sub-Block Inventory

Sub-Block	Domain	Function
Config Registers	MC	All timing parameters, page policy, refresh mode, RFM thresholds, DFI offsets. Written via AXI4-Lite CSR path.
Init FSM	MC	Full DDR5 power-up: RESET_n, NOP burst, DLL reset, ZQcal via MPC, MRW burst, optional training dispatch. Single instance.
Bank State Machines	MC	16 instances, one per bank. Tracks bank state and owns all per-bank timers. Source of truth for bank availability.
Timing Control	MC	Combinational + registered timer array. Produces <code>cmd_ok[15:0][cmd_type]</code> each cycle for the Scheduler.
Arbitration Logic	MC	Picks winner command class each cycle. Priority, age-based anti-starvation, watermark RD/WR mode switching.
Command Selector	MC	Takes winner class, applies timing constraints, enforces page policy, emits <code>{cmd, rank, BG, bank, row, col}</code> .
Refresh Scheduler	MC	Leaky-bucket tREFI tracking. REFab and REFsb. Up to 8× postpone-ment; <code>ref_urgent</code> on overflow.
ZQcal FSM	MC	Periodic MPC ZQCAL Start/Latch sequence.
RFM FSM	MC	DDR5 rowhammer mitigation. Per-bank RAA counters; triggers RFM when $RAA \leq RAAIMT$.
Power Management FSM	MC	CKE-controlled transitions: Active PD, Precharge PD, Self-Refresh, MPSM.
Write Data Path	MC	BL16-aligned write buffer, DM/ECC generator, Write CRC (CRC-8), DFI write timing pipeline.
Read Data Path	MC	Read capture FIFO, Read CRC/ECC checker, read latency counter, ECS FSM.

2 Config Registers

All timing parameters consumed by the MC Core live here. The host programs them before asserting INIT_KICK and may update them at runtime only while the controller is idle. The Scheduler reads timing values combinatorially every cycle.

Register	Width	Description
CL	7b	CAS Latency (nCK). Drives RL in RTW formula.
CWL	7b	CAS Write Latency = CL-2 (fixed in DDR5).
tRCD	8b	RAS-to-CAS delay. ACT to first RD/WR on same bank.
tRP	8b	Row Precharge Time. PRE to next ACT on same bank.
tRAS	9b	Row Active Time minimum. ACT to PRE minimum.
tRC	9b	Row Cycle Time = tRAS+tRP. ACT-to-ACT same bank.
tWR	8b	Write Recovery Time. WR burst end to PRE.
tRTP	6b	Read-to-Precharge. RD cmd to PRE on same bank.
tCCD_L	5b	CAS-to-CAS, same BG (nCK).
tCCD_S	5b	CAS-to-CAS, diff BG = BL/2 = 8 nCK.
tCCD_L_WR	6b	WR-to-WR same BG. max(32 nCK, 20 ns).
tCCD_L_WR2	6b	WR-to-WR same BG, 2nd not RMW. max(16 nCK, 10 ns).
tWTR_L / tWTR_S	6b/5b	Write-to-Read same/diff BG.
tRRD_L / tRRD_S	5b/4b	ACT-to-ACT same/diff BG.
tFAW	6b	Four-Activate Window (nCK).
tRFC1	12b	REFab recovery time. 195 ns for 8 Gb DDR5.
tRFCsb	11b	REFsb per-bank recovery time.
tREFI	14b	Average refresh interval. 3.9 μ s at $\leq 85^{\circ}\text{C}$.
tMRD	5b	MRW-to-MRW minimum. 16 nCK in DDR5.
tXP / tXS	5b/12b	PDX-to-command / SRX-to-command.
tCKSRE / tCKSRX	5b	Stable clock cycles after/before SR entry/exit.
tDLLK	11b	DLL lock time after DLL Reset MPC. 1024 nCK.
RAAIMT / RAAMMT	8b	RFM RAA Initial/Maximum Management Threshold.
RAADec	4b	RAA decrement per REF command issued.
REF_MODE	2b	00=REFab, 01=REFsb, 10=FGR-2 $\times$, 11=FGR-4 $\times$.
PAGE_POLICY	2b	00=Open, 01=Closed, 10=Adaptive.
AGE_THR1 / AGE_THR2	8b	Intra/cross-class age escalation thresholds.
WR_HIGH_WM / WR_LOW_WM	6b	Write pool watermark high/low.
PHY_WRLAT / PHY_RDLAT	6b	DFI write/read latency pipeline offsets.
FREQ_RATIO	2b	DFI frequency ratio: 00=1:1, 01=1:2, 10=1:4.
INIT_KICK	1b	Write 1 to trigger Init FSM.
SOFT_RESET	1b	Synchronous soft-reset of all MC Core FSMs.

3 Init FSM

Reference: JESD79-5 §3.3.1 (power-up with stable power) and §3.3.2 (reset with stable power). Single instance. Triggered by INIT_KICK. Emits DFI commands directly, bypassing the Cmd Scheduler. Asserts `init_done` on completion to release the Scheduler.

3.1 State Diagram

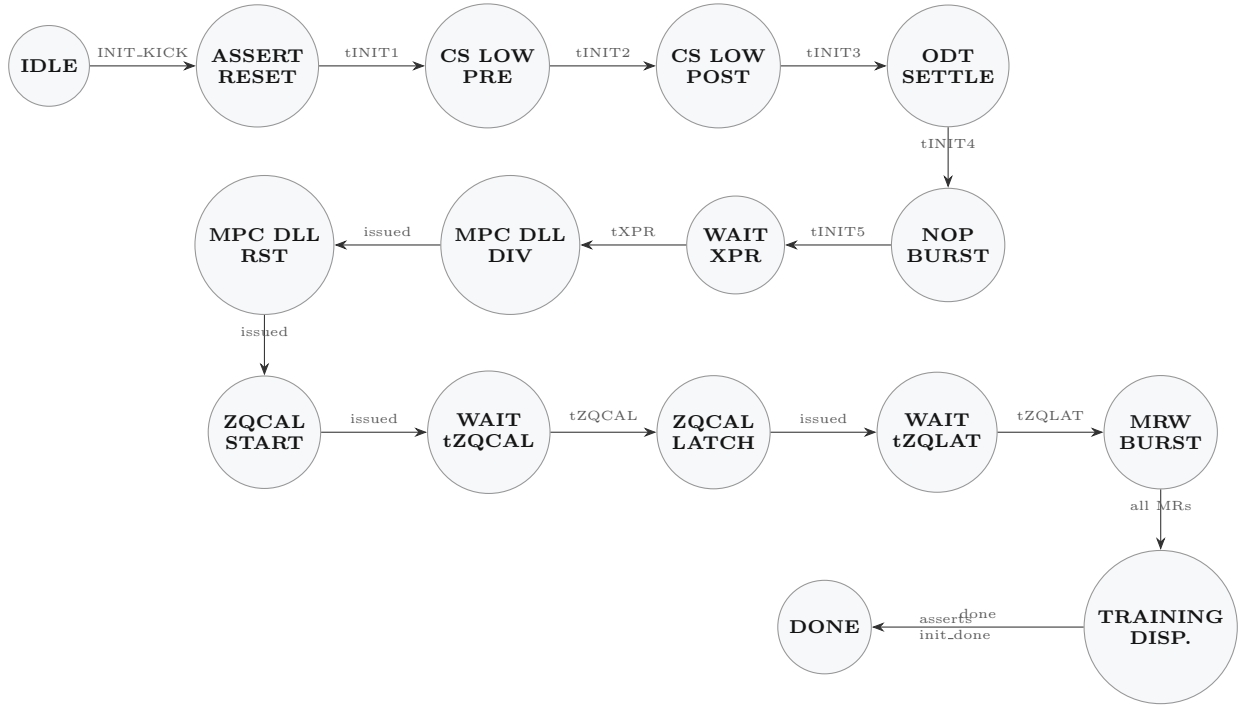


Figure 2. Init FSM state sequence. Executes once on INIT_KICK; idles until SOFT_RESET or re-kick. DFI commands issued directly, bypassing Scheduler.

3.2 State Descriptions

State	Action	Exit Condition
IDLE	Hold RESET _n =0, CKE=0, CS _n =0. All DFI outputs de-asserted.	INIT_KICK=1
ASSERT_RESET	RESET _n held LOW. CS _n LOW. Load tINIT1 (200 μs) counter.	tINIT1 expires
CS_LOW_PRE	CS _n LOW. Count tINIT2 (10 ns) before releasing RESET _n .	tINIT2 expires
CS_LOW_POST	Deassert RESET _n . CS _n remains LOW. CKE LOW. Count tINIT3 (4 ms).	tINIT3 expires
ODT_SETTLE	Raise CKE. Stable CK running. Count tINIT4 (2 μs).	tINIT4 expires
NOP_BURST	Issue ≥3 NOP/DES on DFI. CS _n driven normally.	tINIT5 NOP count
WAIT_XPR	Count tXPR = max(5 nCK, t _{RFC1} + 10 ns).	tXPR expires
MPC_DLL_DIVIDER	Issue MPC: DLL Divider Select (configures tCCD _L divider via MR13).	cmd accepted
MPC_DLL_RESET	Issue MPC: DLL Reset. Required before ZQcal.	cmd accepted
ZQCAL_START	Issue MPC opcode ZQCAL Start (OP=0x00).	cmd accepted
WAIT_tZQCAL	Wait. DRAM performs internal ZQ calibration.	1 μs expires
ZQCAL_LATCH	Issue MPC opcode ZQCAL Latch (OP=0x01).	cmd accepted
WAIT_tZQLAT	Wait tZQLAT = 30 nCK.	tZQLAT expires
MRW_BURST	Issue MRW to MR0, MR2, MR4, MR5, MR6, MR8, MR32–35, MR58. One per tMRD.	all MRWs done
TRAINING_DISPATCH	CA/CS training, Write Leveling, Read training. Controlled by TRAIN_EN bits.	training done
DONE	Assert init_done. Release Cmd Scheduler from reset.	permanent

3.3 MRW Sequence

MR0 (BL, CL), MR2 (2N mode, write leveling), MR4 (refresh/FGR), MR5 (RTT_PARK, DM), MR6 (tWR, tRTP), MR8 (BL, preamble, CRC enable), MR32–35 (ODT), MR58 (RFM: read via MRR, mirror to RAAIMT/RAAMMT config regs). Each write separated by tMRD (16 nCK).

4 Bank State Machines (BSM)

16 instances, one per bank ($2BG \times 4 \text{ banks} \times 2 \text{ sub-channels}$, or $4BG \times 4 \text{ banks}$ for x4/x8 devices). Each BSM is independent, receives commands from the Cmd Scheduler, and owns all per-bank timers.

4.1 State Diagram

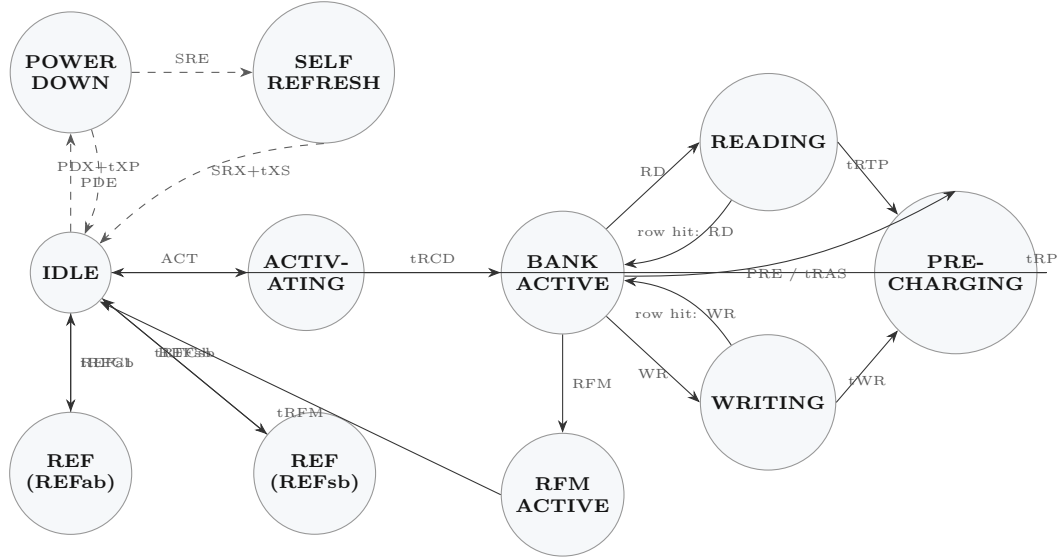


Figure 3. Bank State Machine (single bank). 16 instances run independently. Dashed arrows: power-state transitions (CKE-controlled).

4.2 State Descriptions

State	Legal Inputs	Timer / Exit
IDLE	ACT, REFab, REFsb, PDE, SRE, RFM	No timer. Exits on received command.
ACTIVATING	(internal)	tRCD countdown. Transitions to BANK_ACTIVE when expired.
BANK_ACTIVE	RD, RDA, WR, WRA, PRE, PREA	tRAS loaded at entry. PRE only after tRAS satisfied. Open row registered.
READING	BST (BC8 truncate)	CL+BL/2 until burst end. RDA: hidden PRE starts at tRTP.
WRITING	BST (BC8 truncate)	CWL+BL/2 until burst end. WRA: hidden PRE at CWL+BL/2.
PRECHARGING	—	tRP counter. Transitions to IDLE when expired.
REFRESHING (REFab)	—	tRFC1 counter. All 16 banks enter simultaneously.
REFRESHING (REFsb)	—	tRFCsb counter. Only target bank locked; others remain accessible.
POWER DOWN	PDX	Precharge PD (from IDLE) or Active PD (from ACTIVE). Exits after PDX+tXP.
SELF REFRESH	SRX	tCKSRE after entry. Min tCKESR. SRX: tCKSRX then tXS before commands.
RFM ACTIVE	—	tRFM counter (same duration as tRFC1). Issued by RFM FSM.

4.3 Per-Bank Timer Summary

Timer	Loaded By	Guards
tRCD	ACT issued	First RD/WR on same bank
tRAS	ACT issued	PRE/PREA on same bank (minimum open)
tRC	ACT issued	Next ACT on same bank
tRP	PRE issued	Next ACT on same bank
tWR	WR burst end	PRE on same bank
tRTP	RD cmd issued	PRE on same bank
tRFC1	REFab issued	Any command on any bank
tRFCsb	REFsb issued	Commands to the refreshed bank only
tXS	SRX asserted	First ACT after self-refresh exit
tXP	PDX asserted	First command after power-down exit
tMRD	MRW issued	Next MRW

5 Timing Control

The Timing Control block is not a state machine. It is a combinational and registered timer array that produces `cmd_ok[15:0]` [CMD_TYPES] every MC clock cycle. The Scheduler may only issue a command if the corresponding `cmd_ok` bit is asserted.

5.1 Cross-Bank Constraint Checkers

Constraint	Scope	Implementation
tRRD.L	Same BG, any bank	Timestamp of last ACT in same BG. Block until $T_{now} - T_{last_ACT_BG} \geq t_{RRD.L}$.
tRRD.S	Diff BG, any bank	Timestamp of last ACT globally. Block until $\Delta T \geq t_{RRD.S}$.
tFAW	All banks, same rank	4-entry ring buffer of recent ACT timestamps. Block if oldest is within tFAW.
tCCD.L	Same BG, RD/WR	Timestamp of last CAS in same BG. tCCD.L.WR for WR; tCCD.L.WR2 for non-RMW.
tCCD.S	Diff BG	Timestamp of last CAS globally. $\Delta T \geq t_{CCD.S} = BL/2$.
tWTR.L	Same BG, WR→RD	Last write data end in same BG. Block RD until $\Delta T \geq WL + BL/2 + t_{WTR.L}$.
tWTR.S	Diff BG, WR→RD	Same but using $t_{WTR.S}$.
tRTW	Any, RD→WR	Last RD cmd. Block WR until $\Delta T \geq RL + BL/2 - WL + 2 + t_{WPRE}$.

6 DDR Command Scheduler

The Scheduler is the arbitration heart of the MC Core. It consumes `cmd_ok` from Timing Control, bank states from BSMs, and pending transactions from the RD/WR Cmd Pools. It issues one DFI command per clock cycle.

6.1 Scheduler FSM

6.2 Command Priority Order

1. Refresh Override — REF budget about to expire (highest priority)
2. RFM — $RAA \leq RAA_{MT}$ (safety critical)
3. ZQcal — periodic, low urgency until overdue
4. Write drain — write buffer nearly full
5. Row-hit reads — open page, same row
6. Row-hit writes
7. Row-miss reads — ACT → wait tRCD → RD
8. Row-miss writes
9. PRE — close idle banks if closed-page policy active

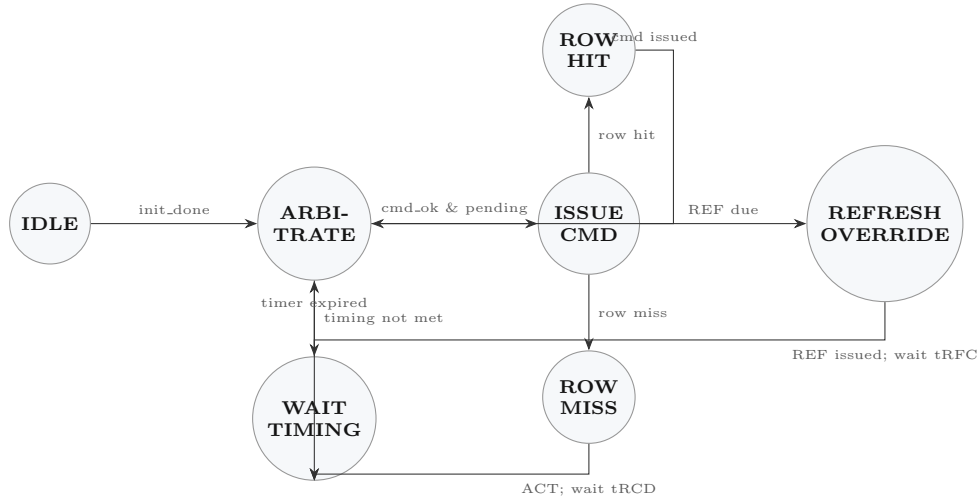


Figure 4. DDR Command Scheduler FSM. One command issued per MC clock cycle.

6.3 Page Policy

Policy	Row Hit	Row Miss	Best For
Open Page	CAS only	PRE $\xrightarrow{t_{RP}}$ ACT $\xrightarrow{t_{RCD}}$ CAS	Sequential, high spatial locality
Closed Page	ACT+CAS (always)	ACT $\xrightarrow{t_{RCD}}$ CAS	Random access; streaming workloads
Adaptive	Hit-rate decides	Sliding window $N = 256$	Mixed; autotunes per bank independently

Adaptive thresholds: $\text{hit_rate} > \text{OPEN_THR}$ (default 70%) $\Rightarrow$ open; $\text{hit_rate} < \text{CLOSE_THR}$ (default 30%) $\Rightarrow$ closed.

6.4 Arbitration Logic

Runs every MC clock. Selects the winning command class. Does not select bank/row/col — that is the Command Selector's job.

Priority hierarchy:

Level	Winner	Condition	Effect
0	REF urgent	$\text{Credits} \geq 8$ or $\Delta t_{\text{since_REF}} > t_{\text{REFI_max}}$	Hard stall; all RD/WR suppressed
1	REF nominal	tREFI countdown reaches zero	REF injected; RD/WR deferred
2	ZQ calibration	tZQCS interval expires	ZQCS after current REF completes
3	Read	RD pool non-empty & RD mode active	BG-aware scheduling
4	Write	WR pool non-empty & WR mode active	Watermark-controlled batching
5	Scrubber/MRR	Background maintenance, idle slot	Pre-emptible

Age counter (anti-starvation): Every RD/WR pool entry carries an age counter incremented each MC cycle. At AGE_THR1 (default 64): intra-class head-of-line bypass. At AGE_THR2 (default 256): WR entry may overtake RD priority. Age resets to 0 on dispatch.

RD/WR mode switching: Switch to WR mode when WR depth $\geq \text{WR_HIGH_WM}$ (32) or RD empty. Switch back when WR depth $\leq \text{WR_LOW_WM}$ (8) or WR empty or urgent RD. Hysteresis gap of 24 entries prevents mode thrashing.

6.5 Command Selector — Legal Check Matrix

All gates must be true before dispatch. Any false $\Rightarrow$ NOP inserted.

Gate	Timer Source	Constraint
bank_avail[b]	BSM state	Bank not in ACTIVATING or PRECHARGING
gate_rcd[b]	ACT issue	t_{RCD} : ACT to first CAS on same bank
gate_rp[b]	PRE issue	t_{RP} : PRE to ACT on same bank
gate_ras[b]	ACT issue	t_{RAS} : ACT to PRE minimum
gate_ccd_l	Same BG CAS	t_{CCD_L} or $t_{CCD_L_WR}$ or $t_{CCD_L_WR2}$
gate_ccd_s	Diff BG CAS	$t_{CCD_S} = BL/2$
gate_rrd_l	Same BG ACT	t_{RRD_L}
gate_rrd_s	Diff BG ACT	t_{RRD_S}
gate_faw	Sliding window	≤ 4 ACTs within t_{FAW}
gate_wtr_l	WR data end, same BG	t_{WTR_L} : WR to RD
gate_wtr_s	WR data end, diff BG	t_{WTR_S} : WR to RD
gate_rtw	RD cmd	$t_{RTW} = RL + BL/2 + 2 - WL$
gate_rfc[r]	REF issue, per rank	t_{RFC1} : all commands blocked
gate_rfcpb[r] [b]	REFsb, per bank	t_{RFCsb} : same bank only blocked
gate_zqcs	ZQCS issue	All commands blocked during ZQ
gate_mrd	MRW issue	t_{MRD} : MRW to MRW

7 Maintenance Engine

Four independent sub-FSMs handle all autonomous background operations. Each posts requests to the Cmd Scheduler via a priority request bus.

7.1 Refresh Scheduler FSM

States: IDLE $\rightarrow$ REF_DUE $\rightarrow$ WAIT_BANKS_IDLE $\rightarrow$ ISSUE_REF $\rightarrow$ WAIT_tRFC $\rightarrow$ DONE $\rightarrow$ IDLE. REFsb path substitutes WAIT_TARGET_BANK_IDLE and WAIT_tRFCsb.

- **Leaky-bucket counter:** increments by 1 each t_{REFI} . Decrements on REF issued. If counter reaches 9: **ref_urgent** asserted, hard-stalling all RD/WR.
- **REFsb cycling:** bank address cycles BA[1:0] in order 00 $\rightarrow$ 01 $\rightarrow$ 10 $\rightarrow$ 11. 8 REFsb commands $\approx$ 1 REFab credit.
- **FGR mode:** REF_MODE[1]=1 $\Rightarrow$ tRFC2; REF_MODE[1:0]=11 $\Rightarrow$ tRFC4. Budget counter threshold halved or quartered.
- **Temperature derating:** MR4 TUF=1 $\Rightarrow$ tREFI halved before loading into credit tracker. MC polls MR4 periodically via background MRR.

7.2 ZQcal FSM

States: IDLE $\rightarrow$ WAIT_IDLE $\rightarrow$ ISSUE_ZQCAL_START $\rightarrow$ WAIT_tZQCAL $\rightarrow$ ISSUE_ZQCAL_LATCH $\rightarrow$ WAIT_tZQLAT $\rightarrow$ DONE $\rightarrow$ IDLE.

Triggered by periodic tZQCS interval timer (CSR-configurable) or host write to ZQCAL_TRIG. All banks must be idle before issuing. During ZQCAL, all commands blocked via **gate_zqcs**. $t_{ZQCAL} = 1 \mu s$; $t_{ZQLAT} = 30 nCK$.

7.3 RFM FSM (Rowhammer Management)

States: IDLE $\rightarrow$ MONITOR_RAA $\rightarrow$ RFM_REQUEST $\rightarrow$ WAIT_ISSUE $\rightarrow$ WAIT_tRFM $\rightarrow$ UPDATE_RAA $\rightarrow$ MONITOR_RAA.

RAA counter logic:

- RAA[b] + = 1 on every ACT to bank *b*.
- RAA[b] – = RAADec on every REF issued to bank *b*.
- RAA[b] $\leq$ RAAIMT: assert RFM request to Scheduler (priority: above ZQcal, below refresh override).
- tRFM = tRFC1 (same duration). Bank enters RFM_ACTIVE during this window.
- After tRFM: RAA[b] reset. Return to MONITOR_RAA.

7.4 Power Management FSM

States: NORMAL $\rightarrow$ PD_ENTRY_CHECK $\rightarrow$ {PRECHARGE_PD or ACTIVE_PD} $\rightarrow$ PDX_WAIT_tXP $\rightarrow$ NORMAL. Self-Refresh branch: NORMAL $\rightarrow$ SR_ENTRY $\rightarrow$ WAIT_tCKSRE $\rightarrow$ SELF_REFRESHING $\rightarrow$ SR_EXIT $\rightarrow$ WAIT_tCKSRX $\rightarrow$ WAIT_tXS $\rightarrow$ WAIT_tDLLK $\rightarrow$ NORMAL.

- **Precharge PD:** from IDLE. All banks idle. CKE deasserted.
- **Active PD:** from BANK_ACTIVE. Bank retains open row. Resumes after PDX+tXP.
- **Self-Refresh:** all banks precharged first. tCKSRE after CKE falls. On SRX: tCKSRX stable clock, then tXS, then tDLLK.
- **MPSM:** MR2 OP[1:0]=01. Deeper than SR. Exit via DES then MRW to clear MPSM.

8 Write Data Path

Sub-Block	Description
Write Data Buffer	BL16-aligned FIFO. Stores write data from CIF Write Cmd Pool via CDC FIFO. Dequeued when Scheduler confirms a write slot.
DM/ECC Generator	Computes per-byte Data Mask. Generates inline Write CRC using CRC-8 polynomial $G(x) = x^4 + x + 1$ over DQ+DM bits per BL16 burst. Write CRC enabled via MR8 OP[6]=1.
WL Align	Inserts programmable pipeline delay = CWL nCK to align DQS preamble to dfi_wrddata_en. Offset from PHY_WRLAT config reg.
DFI Write Timing	Drives dfi_wrddata[127:0], dfi_wrddata_mask[15:0], dfi_wrddata_cs.n, dfi_parity.in. Preamble per MR8 (2tCK default). ODT timing: $t_{ODT_{on}} = WL - 2$, $t_{ODT_{off}} = WL + BL/2$.
Write CRC Error FSM	States: NORMAL $\rightarrow$ CRC_ERROR_DETECTED $\rightarrow$ RETRY_WRITE $\rightarrow$ AUTO_DISABLE. Monitors dfi_alert.n. Retries up to n times (MR51/52). Auto-disables Write CRC via MRW to MR8 if threshold exceeded.

9 Read Data Path

Sub-Block	Description
Read Latency Counter	Per-outstanding-read counter. Tracks CL + read preamble cycles from RD command issue to expected data arrival. Issues dfi_rddata_en exactly PHY_RDLAT cycles before expected dfi_rddata_valid.
Read Data FIFO	Captures dfi_rddata[127:0] when dfi_rddata_valid asserts. Re-aligns per-burst data. Feeds CIF Reorder Buffer with AXI transaction tag.
ECC/CRC Checker	Verifies Read CRC when MR8 OP[7]=1. On error: flags to CIF error register, optionally triggers read retry. Read CRC adds latency adder to RL (JESD79-5 Table 365).
ECS FSM	States: IDLE $\rightarrow$ ECS_ACTIVE $\rightarrow$ WAIT_tECS $\rightarrow$ IDLE. Triggered by MR14 (auto-ECS) or host MPC request. DRAM scrubs on-die ECC during tECS window ($\leq t_{RFC1}$). Error logs in MR16–MR20.

10 DFI 5.2 Interface

DFI timing: dfi_wrddata_en asserted PHY_WRLAT cycles before write data. dfi_rddata_valid arrives PHY_RDLAT cycles after the RD command. Both offsets are PHY-specific and set during DFI training. FREQ_RATIO determines nCK per DFI clock.

Signal	Dir	Description
dfi_address[13:0]	MC→PHY	CA bus. 2-cycle encoding for DDR5 ACT.
dfi_cs_n[1:0]	MC→PHY	Chip select per rank.
dfi_cke[1:0]	MC→PHY	Clock Enable per rank.
dfi_odt[1:0]	MC→PHY	On-Die Termination control per rank.
dfi_reset_n	MC→PHY	DRAM RESET_n. Driven by Init FSM.
dfi_act_n	MC→PHY	Activate indicator (DDR5).
dfi_bank[1:0]	MC→PHY	Bank address.
dfi_bg[2:0]	MC→PHY	Bank group. BG[2:0] for x4/x8; BG[1:0] for x16.
dfi_wrdata[127:0]	MC→PHY	Write data (BL16 × DQ8 per sub-channel).
dfi_wrdata_mask[15:0]	MC→PHY	Write data mask per byte lane.
dfi_wrdata_en	MC→PHY	Asserted PHY_WRLAT cycles early.
dfi_wrdata_cs_n	MC→PHY	Per-DRAM write chip select (DDR5 PDA).
dfi_parity_in	MC→PHY	CA parity bit (DDR5).
dfi_rddata[127:0]	PHY→MC	Captured read data.
dfi_rddata_valid	PHY→MC	Read data valid flag.
dfi_rddata_dnv	PHY→MC	Data-not-valid error flag.
dfi_alert_n	PHY→MC	DRAM alert: Write CRC or CA parity error.
dfi_ctrlupd_req/ack	MC↔PHY	Controller-initiated PHY update handshake.
dfi_phyupd_req/ack	PHY↔MC	PHY-initiated update/pause handshake.
dfi_init_start/complete	MC↔PHY	Init FSM PHY initialization handshake.
dfi_freq_ratio[1:0]	MC→PHY	00=1:1, 01=1:2, 10=1:4.

11 FSM Inventory

FSM	States	Instances	Notes
Init FSM	16	1	RESET_n through training complete
Bank State Machine	11	16	One per bank; 176 total state bits
Arbitration Logic	6	1	RD/WR mode + priority selection
Command Selector	4	1	Page policy + legal check + sequence gen
Refresh Scheduler	7	1	REFab + REFsb + FGR + urgent override
ZQcal FSM	7	1	MPC ZQCAL Start/Latch flow
RFM FSM	6	1	Per-bank RAA monitoring + RFM injection
Power Management FSM	10	1	PD (Pre/Act), SR, MPSM
Write CRC Error FSM	4	1	dfi_alert_n monitoring; retry + auto-disable
ECS FSM	4	1	DDR5 inline ECC scrub window
Total	≈75	23	

12 Timing Reference

Reference values at 200 MHz (tCK=5 ns, CL=3, CWL=1, BL/2=8) and 100 MHz (tCK=10 ns). Production values set via Config Regs from actual speed bin.

Command Sequence	Formula	@200 MHz	@100 MHz
ACT → RD/WR (same bank)	tRCD	4	4
ACT → PRE (same bank, min)	tRAS	7	4
PRE → ACT (same bank)	tRP	4	4
ACT → ACT (same bank)	tRC = tRAS+tRP	11	8
ACT → ACT (same BG)	tRRD.L	4	4
ACT → ACT (diff BG)	tRRD.S	2	2
RD → RD (same BG)	tCCD.L	8	8
WR → WR (same BG)	tCCD.L.WR	32	32
WR → WR same BG (not RMW)	tCCD.L.WR2	16	16
RD/WR (diff BG)	tCCD.S = BL/2	8	8
RD → WR (any)	RL+BL/2−WL+2+tWPRE	14	14
WR → RD (same BG)	WL+BL/2+tWTR.L	13	13
WR → RD (diff BG)	WL+BL/2+tWTR.S	11	11
REFab → any	tRFC1	39	20
SRX → ACT	tXS = tRFC1+10 ns	41	21
ZQCAL Start → Latch	tZQCAL = 1 μs	200	100
4 ACT window	tFAW	16	16

13 DDR5 vs DDR4 — Design Deltas

Feature	DDR4	DDR5 — MC Core Impact
BL	BL8 (BL/2=4)	BL16 (BL/2=8). All RTW, WTR, WR-to-PRE formulae scale by 2.
ACT encoding	1-cycle CA	2-cycle CA. Scheduler drives CA on two consecutive rising edges.
REFsb	Not supported	Refresh Scheduler adds REFsb path; per-bank tRFCsb timer in BSM.
RFM	Not supported	Full RFM FSM; per-bank RAA counters; RAAIMT/RAAMMT in Config Regs.
ZQ mechanism	ZQCL/ZQCS commands	MPC ZQCAL Start+Latch. Init FSM restructured; ZQcal FSM uses MPC.
tRFC1 (8 Gb)	350 ns	195 ns. Shorter penalty; more scheduling headroom.
tREFI	7.8 μs	3.9 μs. Double refresh rate; budget counter threshold halved.
CWL	Separate lookup	CWL = CL−2 (fixed). Simplifies Config Reg and RTW formula.
tMOD	24 nCK after MRS	Replaced by tMRD only (16 nCK). No separate tMOD path.
tCCD.L.WR	Not defined	max(32 nCK, 20 ns). New gate in Command Selector.
Write CRC	Optional (MR5)	Mandatory support. <code>dfi.alert.n</code> monitoring; Write CRC Error FSM.
ECS	Not supported	ECS FSM in Read Data Path; hooks into tRFC window.
MR addressing	3 MRs	256 MRs (MR0–MR255) via MRW/MRR. Init FSM MRW burst extended.

MC Core Design Specification — Memory Controller Capstone Project. JEDEC JESD79-5 (DDR5) / DFI 5.2 / AXI4. Timing values device-specific; validate against datasheet for each speed bin.